

How the HHL Algorithm is Used in Quantum Machine Learning

Linear Systems, Kernels, and Spectral Filtering

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Spring 2026

The Core Idea

The HHL algorithm solves a linear system

$$Ax = b$$

by preparing a quantum state proportional to

$$|x\rangle \propto A^{-1} |b\rangle.$$

In quantum machine learning, this is important because many learning tasks reduce to

- linear systems,
- matrix inversion,
- least-squares problems,
- kernel methods.

Why HHL Appears in Machine Learning

Many machine learning algorithms can be reformulated as

$$Ax = b$$

for a suitable matrix A and vector b .

Examples include:

- linear regression,
- ridge regression,
- kernel methods,
- Gaussian processes,
- linearized neural networks.

Thus HHL serves as a quantum subroutine for solving the underlying learning problem.

Ordinary Least Squares

In linear regression, one minimizes

$$\min_w \|Xw - y\|^2.$$

The formal solution is

$$w = (X^T X)^{-1} X^T y.$$

This can be written as a linear system:

$$Aw = b, \quad A = X^T X, \quad b = X^T y.$$

Instead of returning the vector w explicitly, HHL prepares a quantum state proportional to

$$|w\rangle \propto (X^T X)^{-1} X^T |y\rangle.$$

This means:

- HHL does not directly output all regression coefficients,
- but it gives quantum access to the solution state,
- which can be used to estimate global properties or predictions.

Kernel Ridge Regression and Related Methods

Kernel methods often require solving

$$(K + \lambda I)\alpha = y,$$

where:

- K is the kernel matrix,
- λ is a regularization parameter,
- α determines the classifier or regressor.

This is again a linear system, so HHL can in principle be applied.

Connection to Quantum Support Vector Machines

In a quantum kernel method, one defines

$$K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2,$$

where $|\phi(x)\rangle$ is a quantum feature state.

A more fully quantum workflow would:

- 1 compute the quantum kernel entries on a quantum computer,
- 2 use HHL to solve the linear system involving K ,
- 3 use the resulting coefficients for classification or regression.

Gaussian Process Regression

Gaussian process regression requires solving systems of the form

$$(K + \sigma^2 I)^{-1} y,$$

where:

- K is the covariance or kernel matrix,
- σ^2 is the noise variance.

The predictive mean is typically

$$\mu(x_*) = k_*^T (K + \sigma^2 I)^{-1} y.$$

This again fits naturally into the HHL framework.

Neural Tangent Kernel Perspective

In the linearized regime of neural network training, one approximates

$$f(x) \approx f(x_0) + \nabla_{\theta} f \cdot (\theta - \theta_0).$$

Training then reduces to solving a linear system involving the neural tangent kernel (NTK).

Therefore, HHL could in principle be used as a quantum linear solver in this regime.

HHL as a Spectral Inverse

From a physics perspective, HHL implements

$$A^{-1}.$$

If

$$A = \sum_j \lambda_j |u_j\rangle \langle u_j|,$$

then

$$A^{-1} = \sum_j \frac{1}{\lambda_j} |u_j\rangle \langle u_j|.$$

So HHL acts as a spectral filter:

$$\lambda_j \mapsto \frac{1}{\lambda_j}.$$

Connection to Green's Functions

In many-body physics, inverse operators appear as resolvents or Green's functions:

$$G(z) = (zI - H)^{-1}.$$

This means that HHL can be understood as applying a Green's-function-like operator to data.

From this viewpoint:

- regression corresponds to a response problem,
- kernels resemble propagators,
- learning becomes spectral filtering.

When HHL Can Help

In principle, HHL can offer strong speedups if:

- the matrix is sparse or efficiently block-encoded,
- the condition number is not too large,
- the input state can be prepared efficiently,
- one only needs global properties of the solution.

Important Caveats

There are important limitations:

- HHL outputs a quantum state, not the full classical vector,
- reading out all coefficients may destroy the speedup,
- data loading can be costly,
- conditioning and noise may limit practical use.

Thus HHL is most useful when one wants

- expectation values,
- overlaps,
- predictions,
- or other global observables.

Summary

HHL enters quantum machine learning as a quantum linear solver for problems such as:

- linear regression,
- kernel methods,
- Gaussian processes,
- linearized neural networks.

Conceptually, one may summarize the structure as

machine learning $\longleftrightarrow$ linear systems $\longleftrightarrow$ resolvents $\longleftrightarrow$ HHL.

Thus HHL is best viewed as a foundational inverse-operator subroutine in quantum machine learning.